

CS-C3240 Machine Learning

Predicting Finland’s electricity consumption from the weather and time

Project report

October 6, 2026

1 Introduction

Weather affects how people use electricity. Cold weather increases demand for heating, while hot weather causes air conditioners to run longer. Strong winds draw heat away from structures more quickly, further increasing heating demand. Humidity changes how warm a building feels, which shifts demand for both cooling and heating [1]. Electricity consumption also follows daily and weekly patterns independent of weather, since usage tends to vary by hour of the day and by day of the week [2].

The goal of this project is to predict how weather conditions in Finland, namely temperature, wind, and humidity, along with the hour of day, weekday and month, affect electricity consumption. This is done by combining data from Fingrid, the Finnish electricity transmission system operator, and the Finnish Meteorological Institute (FMI).

Section 2 describes the data and formulates the task as a regression problem. Section 3 explains how the data was prepared and the features chosen, presents the three methods used (polynomial regression, a decision tree and a random forest) together with their loss function, and describes how the data was split for validation and testing. Section 4 compares the training and validation errors of the methods, chooses the final method and reports its test error. Section 5 interprets the results, discusses the limitations of the study and suggests directions for future work.

2 Problem formulation

The data in this study comes from Fingrid and the Finnish Meteorological Institute (FMI). The electricity data is the “Electricity consumption in Finland” dataset from Fingrid’s open data service [3], comprising 119 369 measurements, and the weather data consists of 52 589 hourly observations from the FMI Tapiola weather station, downloaded from FMI’s open data service [4]. Both cover the years 2020–2025.

Fingrid reported consumption hourly before 13 June 2023 and every 15 minutes thereafter, so the Fingrid data was first aggregated to hourly resolution by averaging all measurements within each hour. The two datasets were then merged on the hourly timestamp. 250 hours were discarded because they lacked temperature, wind speed or humidity observations, leaving a final dataset of 52 241 data points.

Each data point represents one hour and has seven numerical features: air temperature in degrees Celsius (`temp_c`), wind speed in metres per second (`wind_ms`), relative humidity as a percentage (`humid_pc`), a cyclic encoding of the hour of day as sine and cosine (`hour_sin`, `hour_cos`), the weekday as an integer from 1 to 7 (`weekday`), and the month as an integer from 1 to 12 (`month`). The hour of day is encoded cyclically so that hour 23 and hour 0 are treated as adjacent rather than maximally distant. The label is the average electricity consumption in Finland during that hour, measured in megawatt-hours (`consumption_mwh`). Since the label is continuous, this is a

supervised regression problem. The first rows of the merged data are shown in Table 1.

time_utc	local_time	consumption_mwh	temp_c	wind_ms	humid_pc	hour	hour_sin	hour_cos	weekday	is_weekend	month
2019-12-31T22:00:00Z	2020-01-01 00:00:00	9822	-1.5	4.3	83	0	0.0000	1.0000	3	0	1
2019-12-31T23:00:00Z	2020-01-01 01:00:00	9548	-1.1	4.3	85	1	0.2588	0.9659	3	0	1
2020-01-01T01:00:00Z	2020-01-01 03:00:00	9141	-0.7	4.5	87	3	0.7071	0.7071	3	0	1
2020-01-01T02:00:00Z	2020-01-01 04:00:00	9149	0.1	5.4	83	4	0.8660	0.5000	3	0	1
2020-01-01T03:00:00Z	2020-01-01 05:00:00	9206	0.5	5.2	82	5	0.9659	0.2588	3	0	1
2020-01-01T04:00:00Z	2020-01-01 06:00:00	9432	0.3	4.7	83	6	1.0000	0.0000	3	0	1
2020-01-01T05:00:00Z	2020-01-01 07:00:00	9419	0.1	4.7	85	7	0.9659	-0.2588	3	0	1
2020-01-01T06:00:00Z	2020-01-01 08:00:00	9558	0.8	4.9	81	8	0.8660	-0.5000	3	0	1
2020-01-01T07:00:00Z	2020-01-01 09:00:00	9522	1.5	4.6	77	9	0.7071	-0.7071	3	0	1

Table 1: The first rows of the merged raw data (`hour_sin` and `hour_cos` rounded to four decimals).

3 Methods

3.1 Preparing the data

We selected temperature, wind, and humidity based on known physical effects on heating and cooling demand, and included hour of day, weekday, and month to capture daily and seasonal consumption patterns. The plots in Figures 1–6 and the correlation matrix in Figure 7 confirm that consumption varies with each of these features: temperature has by far the strongest relationship with consumption, consumption follows a clear daily and seasonal pattern and is lower at weekends, while wind speed and humidity have weaker effects.

We decided to drop the columns `time_utc`, `local_time`, `hour`, and `is_weekend`. Columns `time_utc`, `local_time`, and `hour` are redundant once cyclically encoded as `hour_sin` and `hour_cos`, and raw timestamp values are not directly usable as model features. Column `is_weekend` is redundant given that `weekday` already captures the same information.

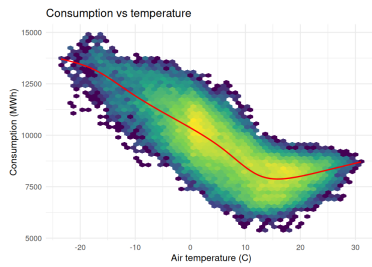


Figure 1: Consumption against air temperature.

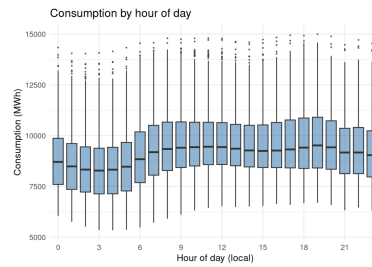


Figure 2: Consumption by hour of day.

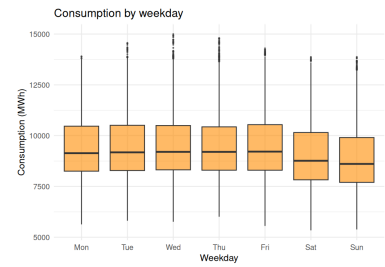


Figure 3: Consumption by weekday.

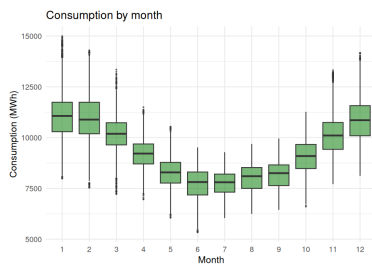


Figure 4: Consumption by month.

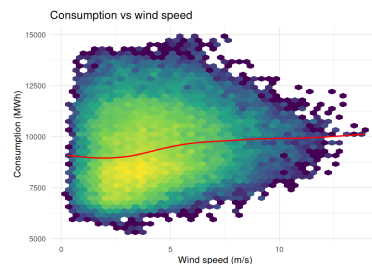


Figure 5: Consumption against wind speed.

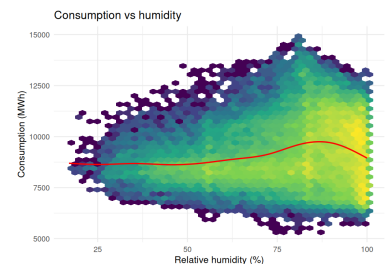


Figure 6: Consumption against relative humidity.

Weekday and month were kept as plain integers rather than cyclically encoded. Unlike hour, where the wraparound between 23:00 and 00:00 occurs within a single day and directly affects short-term consumption patterns, the wraparound effects for weekday and month are less pronounced for this application.

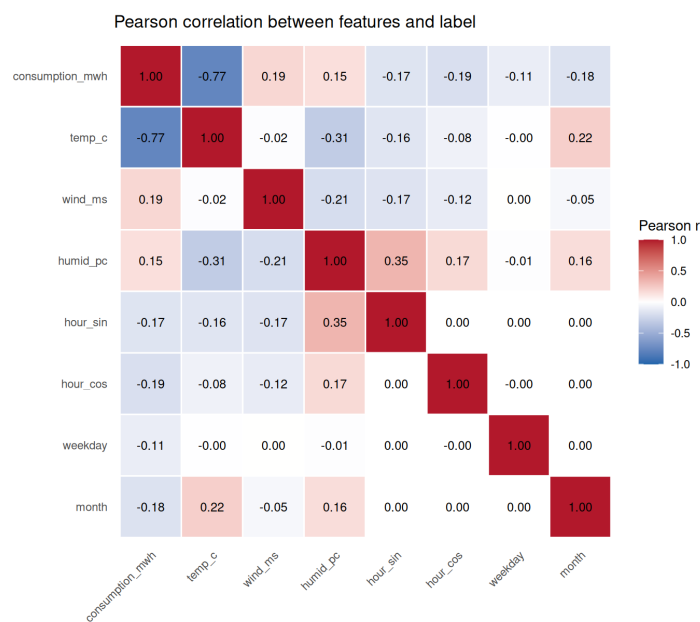


Figure 7: Pearson correlation matrix of the features and the label.

3.2 Models and loss function

Below, a data point is written as $(\mathbf{x}, y)$, where $\mathbf{x} = (x_1, \dots, x_7)^T$ holds the features `temp_c`, `wind_ms`, `humid_pc`, `hour_sin`, `hour_cos`, `weekday` and `month` in this order, and y is the measured consumption `consumption_mwh`. The training set is $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^m$.

Loss function. Since our dataset consists of data points with numeric labels, this is a regression problem, and we chose the squared error loss

$$L((\mathbf{x}, y), h) = (y - h(\mathbf{x}))^2, \quad (1)$$

where $h(\mathbf{x})$ is the predicted consumption. Squared error loss penalizes large prediction errors much more than small ones. This is useful for our problem because large errors in predicted electricity consumption could cause grid operators to over- or under-supply electricity, which is more costly than a small prediction error.

All three models were trained with squared error loss. The polynomial regression minimizes it directly over the training data. The decision tree chooses each split to minimize the squared error within the resulting groups of training points, and predicts the mean consumption of the training points in each leaf. The random forest grows 100 such trees on bootstrap samples with random subsets of features, and predicts the average of the trees' outputs. Using the same loss for all models makes their errors directly comparable.

Polynomial regression. The first model is polynomial regression. Its hypothesis space consists of all polynomials of degree at most d in the seven features. For $d = 2$ this is

$$h(\mathbf{x}) = w_0 + \sum_{j=1}^7 w_j x_j + \sum_{j=1}^7 \sum_{k=j}^7 w_{jk} x_j x_k, \quad (2)$$

which has 36 weights: one intercept, 7 linear terms, 7 squared terms ($j = k$) and 21 pairwise products ($j < k$). The degree d is a hyperparameter. We tried $d = 1$ to 4 and chose the value with the lowest validation error (Section 4.1). The model is linear in its weights, so it is fitted as ordinary least squares on the expanded features, i.e. the weights minimize the average loss (1) on the training set,

$$\min_{\mathbf{w}} \frac{1}{m} \sum_{i=1}^m (y^{(i)} - h(\mathbf{x}^{(i)}))^2. \quad (3)$$

We started from a polynomial model because the correlation matrix (Figure 7) shows a strong linear relationship between temperature and consumption ($r = -0.77$), and because it trains in seconds on about 35 000 rows and has interpretable coefficients. A purely linear predictor ($d = 1$) is not enough, though. The scatter plot in Figure 1 shows that consumption stops falling above roughly 15 °C, which a straight line cannot follow, so squared terms are needed. The product terms let the effect of one feature depend on another: wind should raise heating demand more on a cold day than on a mild one, and the daily cycle may look different in January than in July.

Decision tree. The second model is a decision tree for regression. A tree splits the feature space into rectangular regions $R_1, \dots, R_K$ by asking a sequence of threshold questions on single

features (for example `temp_c` ≤ 2.5), and predicts the average training consumption of the region a data point falls in:

$$h(\mathbf{x}) = \sum_{k=1}^K \hat{y}_k \mathbb{I}\{\mathbf{x} \in R_k\}, \quad \hat{y}_k = \frac{1}{|R_k|} \sum_{i: \mathbf{x}^{(i)} \in R_k} y^{(i)}, \quad (4)$$

where $|R_k|$ is the number of training points in region R_k . The hypothesis space is therefore the set of piecewise constant functions defined by such axis-aligned splits, with the tree limited to a depth of at most D . We tried $D = 2$ to 20 and chose the depth with the lowest validation error, since a tree that is too deep memorises the training data.

We chose a tree because it makes no assumption about the shape of the relationship. It can follow the flattening of the temperature curve, and it finds interactions between features on its own, without us listing them as in the polynomial model. It also suits the weekday and month features, which are stored as integers: a tree can separate weekends from working days or winter from summer with one or two splits, whereas a polynomial has to treat those integers as a numeric scale. Each split is chosen to minimize the squared error within the resulting regions, so the tree is trained with the same loss as the polynomial regression.

Random forest. The third model is a random forest, which predicts the average of $B = 100$ regression trees $h^{(1)}, \dots, h^{(B)}$ of the form (4):

$$h(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B h^{(b)}(\mathbf{x}). \quad (5)$$

Each tree is trained on a bootstrap sample of the training set, drawn with replacement, and at every split only a random subset of m_{feat} features is considered. The hypothesis space is thus the set of averages of 100 such trees. These are still piecewise constant functions, but with far finer steps than a single tree. The trees are not depth-limited; instead each leaf must contain at least n_{min} training points. We tried $m_{\text{feat}} = 1$ to 7 and $n_{\text{min}} \in \{1, 10\}$ and chose the combination with the lowest validation error, keeping the number of trees fixed at 100.

A single tree is unstable: a small change in the training data can change its splits and predictions noticeably. Averaging many trees that have each seen slightly different data and features reduces this variance while keeping the tree's ability to model non-linear effects and interactions. The random forest is the most flexible of our three models, so comparing it with the polynomial regression shows whether that extra flexibility is useful for this problem. The individual trees are trained with squared error, as above.

3.3 Model validation

For model validation the dataset was split chronologically by calendar year. The years 2020–2023 were used for training (34 717 data points, 66.5%), 2024 for validation (8 778 data points, 16.8%) and 2025 for testing (8 746 data points, 16.7%). The test set was set aside and used only once, to evaluate the final chosen model.

We chose a chronological split instead of a random one for two reasons. First, the model is meant to predict future consumption, so the validation and test sets must come from periods it has not seen during training. Second, consumption in consecutive hours is highly correlated, so

a random split would place nearly identical data points in both the training and validation sets and give an over-optimistic error estimate.

Splitting on whole calendar years also ensures that each set covers a complete annual cycle. Had the split been made at an arbitrary point in time, the test set might have contained only summer or only winter months, and the reported error would have reflected a single season rather than year-round performance.

Errors on a set of n data points with predictions $\hat{y}^{(i)} = h(\mathbf{x}^{(i)})$ are reported as the root mean squared error, the mean absolute percentage error and the coefficient of determination:

$$\begin{aligned} \text{RMSE} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2}, & \text{MAPE} &= \frac{100\%}{n} \sum_{i=1}^n \frac{|y^{(i)} - \hat{y}^{(i)}|}{y^{(i)}}, \\ R^2 &= 1 - \frac{\sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2}{\sum_{i=1}^n (y^{(i)} - \bar{y})^2}, \end{aligned} \tag{6}$$

where $\bar{y}$ is the mean consumption of the evaluated set. RMSE is the square root of the average squared error loss (1) and is in MWh, MAPE is the average absolute error as a percentage of actual consumption, and R^2 is the share of the variance in consumption explained by the model, relative to the variance of the evaluated set. Only the squared error loss is used for training, and validation RMSE was used for choosing hyperparameters and the final model.

4 Results

4.1 Training and validation errors

Each model's hyperparameters were chosen by validation error (Section 3.2). For the polynomial regression the validation RMSE was 777, 543, 561 and 563 MWh for degrees 1 to 4, so degree 2 was chosen. The best settings for the other models were maximum depth $D = 9$ for the decision tree, and $m_{\text{feat}} = 4$ with $n_{\text{min}} = 10$ for the random forest. Table 2 shows the resulting errors.

Method	RMSE (MWh)		MAPE		R^2	
	Train	Val	Train	Val	Train	Val
Polynomial regression ($d = 2$)	515	543	4.43 %	4.36 %	0.882	0.900
Decision tree ($D = 9$)	457	619	3.86 %	4.79 %	0.907	0.870
Random forest ($m_{\text{feat}} = 4, n_{\text{min}} = 10$)	351	568	2.94 %	4.42 %	0.945	0.890

Table 2: Training (2020–2023) and validation (2024) errors of the three methods with their best hyperparameters. The best validation value in each column is in bold.

The decision tree has the highest validation error. Its training error is low, but it does not carry over to new data, which is typical for a single deep tree. The random forest has the lowest training error and the largest gap between training and validation error, so it fits the training data more closely than the others, yet its validation error is still higher than that of the polynomial regression.

4.2 Final chosen method

The final chosen method is polynomial regression of degree 2, since it has the lowest validation error on every measure (RMSE 543 MWh, MAPE 4.36 % and $R^2 = 0.900$, compared with 568 MWh, 4.42 % and 0.890 for the random forest). It is also the simplest of the three models, with 36 weights, and its training and validation errors are close to each other (515 vs. 543 MWh), which suggests it does not overfit.

4.3 Test error of the final method

The chosen model was evaluated once on the test set (the year 2025, 8 746 data points). The test set was not used for training or for choosing any setting. The test errors were RMSE = 804 MWh, MAPE = 6.81 % and $R^2 = 0.669$, clearly worse than on the validation set.

Most of this increase comes from a systematic offset. On average the model underestimates 2025 consumption by 625 MWh, about 6.5 % of the mean test-year consumption of 9 652 MWh; the mean over the training years was 9 172 MWh. With this constant offset removed, the test RMSE is 505 MWh, about the same as the training error (515 MWh). The model therefore captures how consumption varies with weather and time of day, but it cannot predict a change in the overall consumption level between years, because none of the features describes such long-term trends. R^2 falls further because it is relative to the variance of the evaluated set (Section 3.3), and consumption varied less in 2025 (standard deviation 1 396 MWh vs. 1 714 MWh in 2024), so the same error is a larger share of the total variation. For comparison, the random forest reached a slightly lower test RMSE (768 MWh), but this was not used in choosing the model.

5 Conclusion

This study compared polynomial regression (degree 2), a decision tree and a random forest for predicting Finland's hourly electricity consumption from weather and time features. All three explained most of the variation on the 2024 validation data (R^2 0.870–0.900). Temperature was by far the strongest predictor ($r = -0.77$), while wind and humidity were much weaker. Polynomial regression of degree 2 was chosen because it had the lowest validation error on every measure (RMSE 543 MWh, MAPE 4.36 %, $R^2 = 0.900$). The tree-based models fit the training data better (random forest training RMSE 351 MWh) but generalised worse, which suggests the relationship is mostly smooth and their extra flexibility brought little benefit.

On the 2025 test set the chosen model reached RMSE = 804 MWh, clearly worse than on validation. Most of the increase was a systematic underestimate of 625 MWh. Without it the test RMSE would be 505 MWh, in line with the training error. The model therefore captures how consumption varies with weather and time, but not changes in the overall consumption level between years. The random forest's slightly lower test RMSE (768 MWh) did not influence the choice, which was made on validation alone.

The results are therefore not optimal, and the study has several limitations. Only one year each was used for validation and testing, so the estimates reflect the weather and demand of those years. No feature describes long-term changes such as electrification, heat pumps or industrial changes. Weather data comes from a single station while consumption is national, and public holidays are not included. The polynomial can also give unrealistic predictions outside the

training range, and a random forest cannot extrapolate at all.

Future work could use rolling-origin validation, training on all past years and testing on the following one, to get error estimates that do not depend on a single year. The features could be extended with weather from several stations and with public holidays, and the model could be given information about the overall consumption level, for example the previous year's average consumption as a feature, or be retrained on the most recent data.

Use of generative AI

Generative AI (Claude, Anthropic) was used to help write the R scripts that draw the plots, to review drafts of this report against the peer-review criteria, to suggest edits to the wording and to format the $\text{\LaTeX}$ document. The data processing, model training, results and interpretation are the authors' own work, and all AI-suggested text was reviewed and edited by the authors.

6 References

- [1] Edge Branding. Weather's effect on energy use behaviour. Montel.energy, 2024. URL <https://montel.energy/resources/blog/how-does-weather-affect-energy-consumption-behaviour>. Accessed: Sept. 16, 2026.
- [2] U.S. Energy Information Administration (EIA). Electricity demand changes in predictable patterns. Today in Energy, 2011. URL <https://www.eia.gov/todayinenergy/detail.php?id=4190>. Accessed: Sept. 16, 2026.
- [3] Fingrid Oyj. Electricity consumption in Finland. Fingrid Open Data, 2026. URL <https://data.fingrid.fi/en/data?datasets=124>. Accessed: Sept. 16, 2026.
- [4] Finnish Meteorological Institute. Download observations. FMI Open Data, 2026. URL <https://en.ilmatieteenlaitos.fi/download-observations>. Accessed: Sept. 16, 2026.

7 Appendix: code

The code for data preparation, plotting and model training is available at <https://anonymous.4open.science/r/ML-project-predicting-consumption-from-weather-FOCB/>.